McMaster University

ENGINEERING
McMaster-Mohawk
Bachelor of Technology
Partnership

SMRTTECH 4ES3 – Embedded Systems

Course Project Introduction:

Fundamentals of **PCB Design**

A. Objective

In this project, you will learn the fundamental concepts behind printed circuit board (PCB) design and design principles required to create reliable and manufacturable PCBs.

This document will help you explore how electronic schematics translate into physical circuit boards, how components are selected and placed, and how electrical signals travel through copper traces on a PCB.

By the end of this document, you will understand:

- The difference between schematic design and PCB layout
- How components and footprints define the physical board
- How power and ground should be distributed
- The structure of a PCB and its layers
- Proper component placement strategies
- Basic trace routing techniques
- The importance of design rules and manufacturability

B. Materials

This lab requires the use of:

- Laptop or Computer
- KiCad 9.0 software
- Project template file provided
- Access to component datasheets

C. Deliverables

Using the schematic template given, create a PCB, route all connections, and fix any errors that may appear from the DRC.

Provide an archived version of your completed PCB and a screenshot.

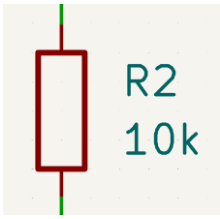
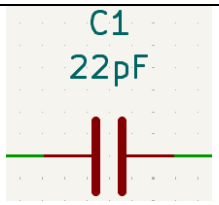
D. Components Recap and Packages

As in the previous lab, engineers start by creating a schematic diagram. A schematic represents the logical electrical connections between components.

A schematic is not concerned with the physical placement of parts. Instead, it shows how signals and power move through the circuit.

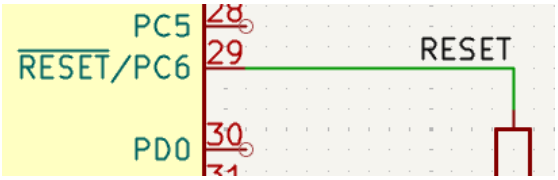
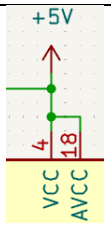
Symbols

Each electronic component is represented by a symbol in the schematic.

Resistor	
Capacitor	

Nets

A net represents an electrical connection between two or more component pins.

Nets can be represented by a label	
Nets can also be represented by symbols	

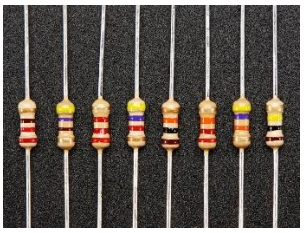
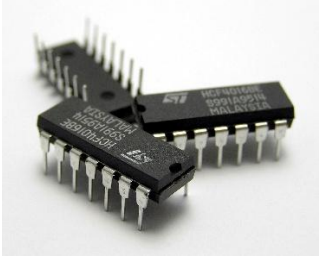
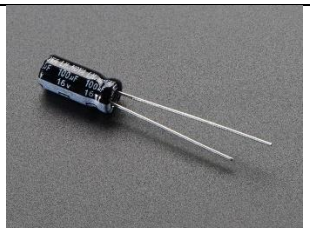
Naming nets helps designers understand signal flow. Examples of net names include:

- VCC
- GND
- LED_SIGNAL
- UART_TX

Each symbol in a schematic must be associated with a **footprint**. A footprint describes the **physical pads and spacing** required for a component to be soldered onto a PCB.

<u>Through-Hole Components (THT)</u>

Through-hole components have leads that pass through drilled holes in the PCB. Examples include:

Through-hole resistors	
DIP integrated circuits	
Large capacitors	

Advantages:

- Easy to solder manually
- Strong mechanical connection

Disadvantages:

- Larger board area
- More expensive for automated assembly

Surface Mount Components (SMD)

Surface mount components sit directly on the surface of the PCB. They are usually soldered directly to exposed pads. Examples include:

Resistors	
QFN microcontrollers	
SOIC packages	

Advantages:

- Smaller components
- Higher density circuits
- Automated manufacturing

Disadvantages:

- Harder to solder manually
- Footprint Sizes

Surface-mount resistors and capacitors come in standard sizes; it is very important to understand the size code to minimize errors while selecting components. Different sizes can handle different power levels and dissipate heat differently.

Codes are often formatted in imperial measurements:

XXYY XX refers to the length, and YY refers to the width.

An 0805 resistor would be approximately 0.08” by 0.05”.

An 0805 resistor corresponds to a 2012 metric resistor.

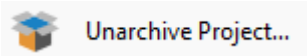
Common Size Table

Imperial				Size	Metric			
Resistor Case Code	Approx. Length (in)	Approx. Width (in)	Power (W)		Resistor Case Code	Approx. Length (mm)	Approx. Width (mm)	Power (W)
01005	0.016	0.008	0.031		**0402**	0.4	0.2	0.031
0201	0.02	0.01	1 / 20 (0.05)		**0603**	0.6	0.3	1 / 20 (0.05)
0402	0.04	0.02	1 / 16 (0.062)		**1005**	1.0	0.5	1 / 16 (0.062)
0603	0.06	0.03	1 / 10 (0.10)		**1608**	1.6	0.8	1 / 10 (0.10)
0805	0.08	0.05	1 / 8 (0.125)		**2012**	2.0	1.25	1 / 8 (0.125)
1206	0.125	0.06	1 / 4 (0.25)		**3216**	3.2	1.6	1 / 4 (0.25)
1210	0.125	0.10	1 / 2 (0.5)		**3225**	3.2	2.5	1 / 2 (0.5)
1812	0.18	0.125	3 / 4 (0.75)		**4532**	4.5	3.2	3 / 4 (0.75)
2010	0.20	0.10	3 / 4 (0.75)		**5025**	5.0	2.5	3 / 4 (0.75)
2512	0.25	0.125	1		**6332**	6.3	3.2	1

Table sourced via allpcb.com

E. Starting your PCB

You will now need to open the lab file provided. Once the ZIP file is downloaded, unarchive the project from the **File** Menu.



Create a new folder or select an existing one. Open that newly unzipped project. Open the Schematic editor and take notes of the Symbols and nets included in this project. Once familiar with the schematic, open the PCB editor. Press **Update PCB from Schematic.**

Once all the footprints load from the schematic, add your edge cuts. Ensure you have selected the correct layer.

Organize your components and connect all nets using the next steps.

F. Power Distribution

Power distribution is one of the most important aspects of PCB design. Electronic components require stable voltage supplies to operate correctly. These nets are often referred to as voltage rails. Most circuits contain several voltage rails, such as:

- 5V
- 3.3V
- GND

These rails distribute power throughout the board, allowing you to power your integrated circuits and other passive components.

<u>Decoupling Capacitors</u>

Decoupling capacitors are used to stabilize voltage around your integrated circuits. When an IC powers up, it can cause a drop in voltage. If that drop is too large, you may have unintended consequences such as:

- Brownout or soft resetting
- Increased Noise or EMI
- Circuit Failure

A decoupling capacitor acts as a local energy reservoir that provides current when the IC switches rapidly. In the provided template, the decoupling capacitors connected to the microcontroller's VCC rail are C1 and C2.

Key rule: Every integrated circuit should have at least one decoupling capacitor placed close to its power pins.

Typical value: 0.1 μF - 1 μF ceramic capacitor

Decoupling Capacitors are often paired with bulk capacitors, often placed further away, to help stabilize the overall power supply. They help reduce voltage drops during large current changes. In the provided template, the bulk capacitor can be seen as C5

Typical values: 10 μF – 100 μF capacitor

G. PCB Layers

A PCB is constructed from several layers that serve different purposes. Layers can include solder mask, copper layers, silkscreen and dielectric.

The **solder mask** is a protective coating that prevents solder bridges and protects copper traces. Historically solder masks were mainly green. Solder masks can be ordered in any colour. Other common colours include black, red, blue, and purple.

The copper layers carry electrical signals. These are the physical traces you set up. Boards can have more than one copper layer. Although this project will focus on two-layer PCBs, PCBs can have up to 32 layers for ultra high-density PCBs.

The Silkscreen is the printed layer used for component labels and markings. Examples include:

- Component reference designators (R1, C1)
- Pin labels
- Logos
- Edge Cuts
- Custom art / Pictures

The dielectric is the core material. The most widely used and common material is FR4 (Flame Retardant 4), which is a fiberglass and epoxy resin binder. Flexible PCBs have become more popular in recent years, with demand for smaller, lighter devices and the rise of wearable devices.

H. Component Placement

Component placement greatly influences the performance and manufacturability of a PCB. Good placement simplifies routing and improves signal integrity.

Functional Grouping

Components that work together should be placed close together. For example, microcontrollers should be placed with their decoupling capacitors, oscillator, and programming header.

Connector Placement

Connectors are usually placed at the edges of the board for easy access. For example, a USB port should be placed at the edge of the board to allow the cable to connect without being blocked.

I. Routing Fundamentals

Routing is the process of connecting component pads using copper traces.

Trace width determines how much current a trace can safely carry. Power traces should generally be wider than signal traces.

KiCad has a built-in trace width calculator under Calculator Tools in the Project file menu.

General guidelines:

Current	Trace Width
100 mA	0.25 mm
500 mA	0.5 mm
1 A	1 mm

Vias:

When traces cannot be routed around other components, Vias may be used. A Via is a conductive hole that travels through the layers of the PCB and creates a connection between sides.

In KiCad, Vias can be placed from the Right-hand menu, and must be placed onto an existing net or pad. The net can be changed through the property menu.

J. Grounding Concepts

Ground is the reference point for all voltages in the circuit. A poorly designed ground system can cause noise and instability.

Ground Plane

A ground plane is a large copper area connected to ground.

Benefits:

- Low-resistance return paths
- Reduced electromagnetic noise
- Improved thermal performance

In KiCad, a ground plane can be created with the **Draw Filled Zone** tool, selecting a point, then select your GND plane, and creating a box around your PCB. Pressing B will fill in the empty area.

K. Design Rules

Manufacturers impose limits on PCB features. These limits are called design rules.

In the previous lab, we used the default Design Rules, in the template provided the JLCPCB Design rules are included.

To complete this introduction, please archive the project files of your completed PCB. Upload the ZIP and a screenshot of the PCB to the Avenue drop box.